Classical problems

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Part I
Problems

1 Stars and Bars (integer compositions)

Link: `https://en.wikipedia.org/wiki/Stars_and_bars_(combinatorics)`

1.1 Use/When

It can be used to solve a variety of **counting problems**, such as **how many ways there are to put n indistinguishable balls into k distinguishable bins**.

$$x_1 + \cdots + x_n = k$$

equals to

- Positive ($x_i > 0$): $\binom{k-1}{n-1}$.

- Nonnegative ($x_i \geq 0$): $\binom{k+n-1}{n-1}$.
- With bounds $x_i \leq b$: use inclusion–exclusion on events $x_i \geq b+1$:

$$\sum_{j=0}^{\lfloor k/(b+1) \rfloor} (-1)^j \binom{n}{j} \binom{k - j(b+1) + n - 1}{n-1}.$$

1.2 Sketch of proof

Count the ways to arrange n stars and k bars together. The constraint $x_i > 0$ is equivalent to there is no bars adjcent and can be put at two nodes Arrange k stars; place $n-1$ bars to split into n piles. Choosing bar positions bijects to solutions.
Example.

$$x_1 + x_2 + x_3 = 7, x_i \geq 0 \Rightarrow \binom{9}{2} = 36.$$

2 Pigeonhole Principle

Link: `https://en.wikipedia.org/wiki/Pigeonhole_principle`

2.1 Use/When.

In mathematics, the pigeonhole principle states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item

Statement. N objects in n boxes $\Rightarrow$ some box has $\geq \lceil N/n \rceil$ objects.
Example. 13 integers $\Rightarrow$ two share a remainder mod 12.

3 Catalan Numbers

Link `https://en.wikipedia.org/wiki/Catalan_number`

3.1 Definition

The Catalan numbers are a sequence of natural numbers that occur in various counting problems, often involving recursively defined objects. These numbers are defined by $C_n = \dfrac{1}{n+1}\binom{2n}{n} = \frac{2n!}{n!(n+1)!}$. As the consequence from the definition,

$$C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}, \quad \text{asymptotic} \quad C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}}.$$

3.2 Examples

Dyck paths A Dyck path is a staircase walk from $(0,0)$ to (n,n) that lies strictly below (but may touch) the diagonal $y = x$. In the other words, we want to move from $(0,0)$ to (n,n) under the condition that each step we can move 1 step in either x or y

direction but the value of x-coordinate must be always less or equal to the value of y-coordinate. **The number of Dyck paths of order n is C_n counts the number of expressions containing n pairs of parentheses which are correctly matched:**

Balanced parentheses C_n counts the number of expressions containing n pairs of parentheses which are correctly matched:

$$n = 2 : ()(), (()); \quad n = 3 : ()()(), (())(), ()(()), ((())), (()()).$$

Binary tree shapes C_n is the number of full binary trees with $n + 1$ leaves, or, equivalently, with a total of n internal nodes, e.g. for $n = 3$

Non three-term patterns C_n is the number of permutations of $1, ..., n$ that avoid the permutation pattern 123 (or, alternatively, **any of the other patterns of length** 3); e.g., the number of permutations with no three-term increasing subsequence. For $n = 4$, permutations with no three term increasing subsequence are

$$1432, 2143, 2413, 2431, 3142, 3214, 3241, 3412, 3421, 4132, 4213, 4231, 4312, 4321.$$

4 Bertrand's Ballot Problem

Link `https://en.wikipedia.org/wiki/Bertrand%27s_ballot_theorem`

4.1 Use/When.

In an election, candidate A receives a votes and candidate B receives b votes with $a > b$. **When** votes are counted in a randomly picked order, the probability that A will (strictly) ahead of B is

- Strictly ahead: $\dfrac{a-b}{a+b}$
- Ahead (ties allowed): $\dfrac{a-b+1}{a+1}$

4.2 Sketch of proof (by reflection)

Any sequence that begins with a vote for B must reach a tie at some point, because A eventually wins overall. The probability that a sequence begins with B is

$$\mathbb{P}(\text{sequence begins with } B) = \frac{b}{a+b}.$$

that begins with A and reaches a tie, reflect the votes up to the point of the first tie (so any A becomes a B, and vice versa) to obtain a sequence that begins with B. Hence, **every sequence that begins with A and reaches a tie is in one-to-one correspondence with a sequence that begins with B**

$$\mathbb{P}(\text{sequence ties and start with } A) = \mathbb{P}(\text{sequence begins with } B).$$

Therefore, the probability that A always leads the vote is

$$\begin{aligned}
\mathbb{P}(A \text{ always leads}) &= 1 - \mathbb{P}(\text{sequence ties at some point}) \\
&= 1 - \mathbb{P}(\text{sequence ties and begins with A or B}) \\
&= 1 - 2\,\mathbb{P}(\text{sequence ties and begins with B}) \\
&= 1 - 2\,\mathbb{P}(\text{sequence begins with B}) \\
&= 1 - 2 \cdot \frac{b}{a+b} \\
&= \frac{a-b}{a+b}.
\end{aligned}$$

4.3 Equivalent problems

Lattice Path Formulation Consider lattice paths from $(0,0)$ to (p,q) using steps:

$$(1,0) \quad (\text{vote for } A), \qquad (0,1) \quad (\text{vote for } B).$$

Then the ballot condition is equivalent to requiring that the path satisfies

$$y < x \quad \text{at all intermediate points.}$$

Random Walk Formulation Let $X_1, \dots, X_{a+b} \in \{+1, -1\}$ with exactly a values equal to $+1$ and b equal to -1. Define the partial sums

$$S_n = \sum_{i=1}^{n} X_i.$$

Then the event that A is always ahead is equivalent to

$$S_n > 0 \quad \text{for all } n = 1, \dots, a+b.$$

Queuing Interpretation Suppose a queue evolves in discrete time:

- arrival: $+1$ customer (vote for A),
- service: -1 customer (vote for B),

with total arrivals a and services b. Then the theorem gives **the probability that the queue length never becomes zero again after time** 1.

Gambler's Ruin Interpretation Let a gambler start with capital 1.

- a times win $+1$
- b times lose -1

Conditioned on ending with capital $a - b > 0$, the theorem gives **the probability that the gambler never goes bankrupt**.

Order Statistics Interpretation Place p points of type A and q points of type B uniformly on $[0, 1]$. Order them increasingly. The ballot condition is equivalent to:

$$\text{the } k\text{-th } A \text{ occurs before the } k\text{-th } B \text{ for all } k.$$

5 Balls-into-Bins (Occupancy)

Link `https://en.wikipedia.org/wiki/Balls_into_bins_problem`

5.1 Use/When.

Throw m balls into n bins uniformly/independently.The expected number of bins with at least one ball inside is

$$\mathbb{E}[\#\text{nonempty}] = n\left[1 - \left(1 - \frac{1}{n}\right)^m\right].$$

Proof. For each bin, the probability that one ball does not go into this bin is $\frac{n-1}{n}$. So the probability that all m balls are not in this bin is $\left(\frac{n-1}{n}\right)^m$. Thus, we deduce the expectation of one bin that is not empty and multiply this with n to get the target expectation.

In addition, the expected number of empty bins is

$$\mathbb{E}[\#\text{nonempty}] = n\left(1 - \frac{1}{n}\right)^m \underset{n\ large}{\approx} ne^{-m/n}.$$

So for large n, empty bin counts $\approx \text{Poisson}(\lambda = m/n)$.

Example. Elevator: 12 riders, 10 floors $\Rightarrow 10(1 - (9/10)^{12}) \approx 7.176$ stops.

6 Birthday Paradox

Link `https://en.wikipedia.org/wiki/Birthday_problem`

6.1 Use/When.

Collision among k samples from n equally likely types.

$$\mathbb{P}(\text{no collision}) = \frac{(n)_k}{n^k} = \prod_{i=0}^{k-1}\left(1 - \frac{i}{n}\right)$$

So

$$\mathbb{P}(\text{collision}) = 1 - \prod_{i=0}^{k-1}\left(1 - \frac{i}{n}\right).$$

For $n \gg i$, $1 - \frac{i}{n} \approx e^i$. Hence, for $n \gg k$

$$\mathbb{P}(\text{collision}) = 1 - \prod_{i=0}^{k-1} e^i = 1 - e^{\sum_{i=0}^{k-1} i} = 1 - e^{\frac{k(k-1)}{2n}}.$$

Approx. $\mathbb{P}(\text{collision}) \approx 1 - \exp(-\dfrac{k(k-1)}{2n})$. Threshold $k \approx \sqrt{2n \ln 2}$.
Example. $n = 365, k = 23 \Rightarrow$ collision ≈ 0.507.

7 Coupon Collector

Link `https://en.wikipedia.org/wiki/Coupon_collector%27s_problem`

7.1 Use/When.

Time to see all n types at least once.
Mean. $\mathbb{E}[T] = nH_n$, $H_n = \sum_{i=1}^n 1/i$. (Stages: $n/(n-i)$, sum over $i = 0..n-1$.)
Asymptotic. $T = n \log n + \gamma n + o(n)$ with Gumbel fluctuations.
Example. $n = 6$ faces $\Rightarrow \mathbb{E}[T] \approx 14.7$.

8 Derangements (Hat-Check)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Permutations with no fixed point.
Count. $!n = n! \sum_{k=0}^n (-1)^k/k!$; hence $\mathbb{P}(\text{no fixed point}) \to 1/e$. Also $\mathbb{E}[\#\text{fixed points}] = 1$.
Sketch. PIE on events $A_i = \{\pi(i) = i\}$.
Example. $!5 = 44$, prob $44/120 \approx 0.3667$.

9 Hypergeometric (without replacement)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Population N with K successes; draw n; X successes.
PMF. $\mathbb{P}(X = x) = \dfrac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$. Mean nK/N; variance $n(K/N)(1 - K/N)\frac{N-n}{N-1}$.
Example. 5 cards, exactly 2 aces: $\binom{4}{2}\binom{48}{3}/\binom{52}{5}$.

10 Negative Hypergeometric (draw until r successes)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Without replacement, draw until r-th success among K in N.
Mean draws. $\mathbb{E}[T] = \dfrac{r(N+1)}{K+1}$. (Order-statistic symmetry of success positions.)
Example. Draws to first ace: $(52 + 1)/(4 + 1) = 10.6$.

11 Geometric & Negative Binomial (with replacement)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Trials with success probability p.

Facts. Geometric: $\mathbb{P}(T = k) = (1-p)^{k-1}p$, $\mathbb{E}[T] = 1/p$, memoryless. Negative binomial: time to r successes, $\mathbb{E}[T] = r/p$, $\mathrm{Var}[T] = r(1-p)/p^2$.

Example. $p = 0.3 \Rightarrow \mathbb{E}[T_{\text{first}}] \approx 3.\overline{3}$; to 5 successes ≈ 16.67.

12 Runs of Heads/Tails (longest run)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Typical/maximum consecutive run in n flips.

Heuristic. $\mathbb{P}(\text{a run of } \geq r) \leq n \cdot 2^{-r}$ (union bound) $\Rightarrow$ longest run $\approx \log_2 n$.

Example. $n = 100 \Rightarrow$ longest run around 6–7.

13 Penney's Game (pattern vs. counter-pattern)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Which length-m pattern appears first in i.i.d. fair flips; nontransitive strategies.

Facts. Expected waiting depends on overlaps; average 2^m. For $m = 3$, best reply to abc is $\bar{b}ab$, win prob 2/3.

Example. Opponent HHT $\Rightarrow$ best THH.

:pmm

14 Gambler's Ruin (1D random walk)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Hitting prob/time on $\{0, 1, \ldots, N\}$; step $+1$ with prob p, -1 with q.

Results. $\mathbb{P}(\text{hit } N \mid i) = \frac{1-(q/p)^i}{1-(q/p)^N}$ for $p \neq q$; fair case $= i/N$. Fair expected time $= i(N-i)$.

Sketch. Solve second-order difference equations or use martingales/optional stopping.

15 Reflection Principle (lattice/Brownian paths)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Events like "ever cross a level"; links ballot & Catalan; maxima of walks.

Sketch. Reflect the path at first crossing to biject violating paths to an easy-count set.

Example. For a symmetric walk S_k, classic identities for $\mathbb{P}(\max_{k \leq n} S_k \geq a)$.

16 Absorbing Markov Chains

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Expected time to absorption; absorption probabilities in finite chains.

Setup. Reorder $P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$. Fundamental matrix $N = (I - Q)^{-1} = \sum_{k \geq 0} Q^k$.

Results. Expected steps $t = N\mathbf{1}$. Absorption matrix $B = NR$.

Example. "Collect-all" problems can be cast with small Q blocks.

17 Records (running maxima)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. How many new highs in i.i.d. continuous samples.

Facts. $\mathbb{P}(i \text{ is a record}) = 1/i$; $\mathbb{E}[\#\text{records}] = H_n$ by indicators & symmetry.

Example. $n = 100 \Rightarrow \mathbb{E} \approx 5.187$.

18 Fixed Points in a Random Permutation

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Matches $\pi(i) = i$.

Facts. $\mathbb{E}[\#\text{fixed points}] = 1$; $\mathbb{P}(\text{none}) \to 1/e$. Derangements via PIE.

Sketch. Indicator method: $\mathbb{E}[\sum I_i] = \sum \mathbb{E}[I_i] = 1$.

19 First-Appearance Order Symmetry

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Order in which distinct types first appear in i.i.d. sampling.

Fact. The order is a uniform random permutation. Blocks A before B with prob $1/\binom{|A|+|B|}{|A|}$.

Example. Die: "all evens before any odd" $= 1/\binom{6}{3} = 1/20$.

20 Buffon's Needle (geometric probability)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Needle of length ℓ dropped on parallel lines distance d apart ($\ell \leq d$).

Result. $\mathbb{P}(\text{cross}) = \dfrac{2\ell}{\pi d}$.

Sketch. Integrate over angle θ and center offset u: cross if $u \leq (\ell/2)\sin\theta$.

21 Bertrand's Paradox (random chord)

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Demonstrates model dependence in "uniform chord" selection.

Fact. "Chord longer than triangle side" is $1/3, 1/2$, or $1/4$ depending on sampling

scheme (endpoints / angle-distance / uniform midpoint).
Moral. Always specify the randomness model.

22 Broken-Stick Problem

Link `https://en.wikipedia.org/wiki/Catalan_number`

Use/When. Two uniform cuts on $[0, 1]$; lengths X, Y, Z with $X + Y + Z = 1$.
Triangle probability. $\mathbb{P}(\text{triangle}) = 1/4$ (all three $< 1/2$ in the simplex).
Extras. $\mathbb{E}[\max] = 11/18,\ \mathbb{E}[\text{mid}] = 5/18,\ \mathbb{E}[\min] = 1/9.$

Further Reading (reliable):

- *Wikipedia:* Stars and Bars; Inclusion–Exclusion; Pigeonhole; Catalan Numbers; Bertrand's Ballot; Hook-length Formula; Occupancy Problem; Birthday Problem; Coupon Collector; Derangements; Hypergeometric/Negative Hypergeometric; Geometric/Negative Binomial; Penney's Game; Gambler's Ruin; Reflection Principle; Absorbing Markov Chain; Records; Buffon's Needle; Bertrand's Paradox; Broken-stick problem.

- Grimaldi, *Discrete and Combinatorial Mathematics.* Feller, *An Introduction to Probability Theory.*

Part II
Useful tools

23 Inclusion–Exclusion Principle (PIE)

Link `https://en.wikipedia.org/wiki/Inclusion%E2%80%93exclusion_principle`

23.1 Use/When.

The inclusion–exclusion principle is a counting technique which generalizes the familiar method of obtaining the **number of elements in the union of two finite sets**; symbolically expressed as

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

In its general formula, the principle of inclusion–exclusion states that for finite sets $A_1, \ldots, A_n$, one has the identity

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n}^{n} |A_i \cap A_j| + (-1)^n \sum_{1 \le i_1 < \cdots < i_{n-1} \le n} \left|\bigcap_{j=i_1}^{i_{n-1}} A_j\right| + (-1)^{n+1} \left|\bigcap_{i=1}^{n} A_i\right|.$$

24 Hook-Length Formula (Young tableaux)

Use/When. # of standard Young tableaux of shape $\lambda \vdash n$.

Statement. $f^\lambda = \dfrac{n!}{\prod_{c\in\lambda} h(c)}$, where $h(c)$ is the hook length of cell c.

Sketch. Greene–Nijenhuis–Wilf hook walk: product of hook reciprocals equals a valid fill probability.

Example. Shape $(3, 2)$: hook lengths $4, 3, 1; 2, 1 \Rightarrow f = 5!/24 = 5$.